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Plug-and-Play Incomplete Multi-View Clustering via Janus-Faced Affinity Learning with Topology Harmonization

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Abstract

Prevailing incomplete multi-view clustering (IMVC) approaches typically fail to account for the interference of view-exclusive artifacts when learning view-consensus representations, which could compromise the fidelity of the resulting similarity measure. Moreover, inconsistencies in anchor order across views may distort the graph structure, impairing the clustering performance. The reliance on carefully-tuned regularization hyper-parameters also usually undermines the model's practical utility. To alleviate these issues, we propose a plug-and-play IMVC framework named PJFTH that incorporates Janus-faced affinity learning with topology harmonization. It explicitly models the exclusive-to-consensus interplay, derives a view-private graph from each view, and adaptively integrates them into a global consensus affinity according to the respective view's intrinsic characteristics. Furthermore, a permutation transformation with unary encoding constraints is applied to anchor matrix, realigning anchor topology while preserving the values. This process synchronizes anchor order prior to similarity integration and maintains original anchor properties. Notably, all components are coupled seamlessly and optimized in a joint manner. Also, the provable overall linear complexity further enlarges its scalability and practicality. Experimental results confirm that PJFTH receives competitive performance compared to several leading methods.

1. Introduction

The rapid evolution of information technology has significantly facilitated the acquisition of data from heterogeneous sources or modalities, giving rise to what is commonly referred to as multi-view data [6, 8, 21, 23, 24, 28, 36, 42, 55]. Clustering has established itself as an effective methodology for analyzing such data, enabling the discovery of potential structures and yielding interpretable insights [2, 5, 9, 17, 19, 25, 45, 52–54]. Nevertheless, in many real-world scenarios, factors such as sensor malfunctions, transmission inaccuracies, or collection omissions frequently result in partial data absence across one or more views, accordingly inducing the challenging problem known as incomplete multi-view clustering (IMVC) [11, 13, 34, 44, 46, 51]. Effectively addressing IMVC requires not only leveraging the available multi-view information but also mitigating the potential negative effects caused by missing views, rendering it an active and significant research topic in machine learning and data mining [12, 13, 15, 18, 41, 43, 50].

In pursuit of effective solutions for IMVC, a variety of methods have been developed in recent years [3, 4, 27, 30, 38, 49]. For example, Gu *et al.* [7] introduce an enhanced dictionary learning mechanism to reconstruct missing entries, and employ a Gaussian error rank criterion to facilitate the extraction of representations embedded in the tensor structure. In contrast to local pairwise sample interactions, Wen *et al.* [31] leverage a confidence-aware graph to model the underlying group-wise structure, and directly learn a unified relationship matrix that serves as a consistent clustering target. Xu *et al.* [39] investigate cluster patterns through mutual information maximization to strengthen feature discrimination, and simultaneously decrease distribution divergence to improve feature correspondence. While these approaches have advanced the field from various perspectives, they share several common limitations. A primary shortcoming is their widespread neglect of the interference caused by view-exclusive artifacts during the consensus representation learning stage, which could pollute the similarity measure. In addition, inconsistencies in anchor order across different views tend to disrupt the graph-building process, adversely affecting the clustering performance. Moreover, the practical deployment of these models is often hindered by a reliance on carefully calibrated regularization hyper-parameters.

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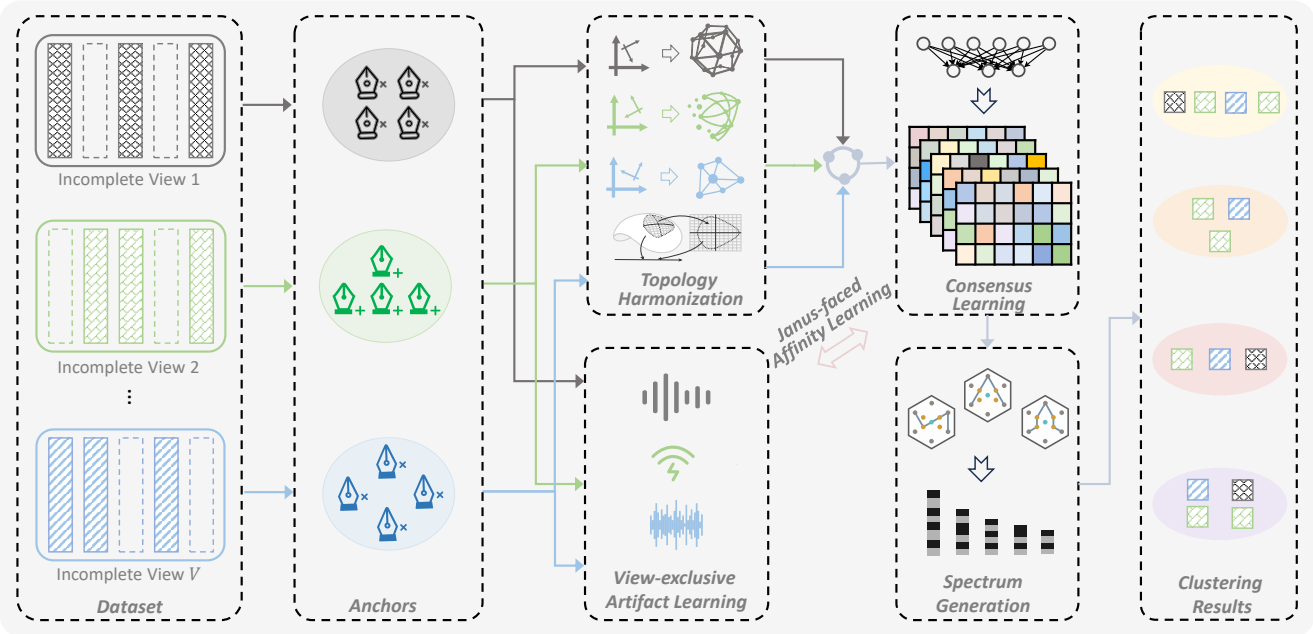


Figure 1. Schematic diagram of PJFTH. It employs a Janus-faced affinity learning paradigm to explicitly characterize and exclude view-exclusive artifacts during consensus representation learning, while utilizing a topology harmonization mechanism to synchronize anchor order across views and thereby provide consistent structures. All components are jointly learned in an end-to-end, non-parametric manner.

To alleviate these limitations, we introduce Janus-faced affinity learning with topology harmonization in this work, and develop a plug-and-play IMVC method named PJFTH. Fig. 1 presents its overall schematic diagram. To be specific, inspired by view-wise learning paradigm, we explicitly characterize view-exclusive artifacts through affinity graph learning applied to each individual view, which is then adaptively incorporated into the consensus learning process according to view itself intrinsic characteristics. This Janus-faced design encourages to strip away and thereby suppress the interference of the respective view's unfavorable factors (like biases induced by incompleteness, signals irrelevant to consensus semantics, etc.) during the consensus affinity generation. Subsequently, given that inconsistencies in anchor order is caused by the disorganized anchor topology, we build a permutation transformation equipped with unary encoding constraints to harmonize the anchor topology. This mechanism aims at synchronizing anchor order prior to similarity integration and meanwhile preserving the originally learned anchor values. Notably, our proposed method operates without involving any regularization hyper-parameters, making it readily applicable as a plug-and-play module in various scenarios. In addition to these, all components are jointly optimized through a six-step iterative updating rule derived for the objective function, promoting mutually reinforcing interactions among modules. Besides, the resulting algorithm exhibits linear complexity, further enhancing its scalability and practical utility. Then, to validate its effectiveness, we conduct extensive comparative experiments on multiple benchmark datasets with varying incompleteness percentages. Results demonstrate that PJFTH achieves competitive performance across most evaluation scenarios. In short, the contributions of this work are as follows,

- We develop an IMVC method, PJFTH, which incorporates Janus-faced affinity learning with topology harmonization into a unified, plug-and-play framework. It separates view-exclusive artifacts from consensus representation learning, while synchronizing anchor order across views, all without requiring any hyper-parameters.
- We derive an efficient six-step iterative updating rule with a linear complexity, further boosting the method's scalability and practical utility in real-world applications.
- We organize extensive experiments on several datasets under a range of incompleteness ratio settings to validate the effectiveness and superiority of PJFTH.

2. Related Work

In recent years, a series of algorithms have been carefully developed for IMVC, progressively advancing the field from various methodological angles. For instance, Wang *et al.* [29] exploit consistent subspace representation via a self-expression layer regularized by $\ell_{1,2}$-norm to improve intra-cluster cohesion. Lin *et al.* [20] jointly conduct feature inference and tensor re-coupling by a t-SVD-based nu-

clear norm to extract high-order view correlations. To enhance robustness, Qin *et al.* [26] model missing patterns using an assumed probabilistic distribution, and adopt an explicit quality calibration to mitigate the inherent biases introduced by varying data quality across samples. Likewise, Yuan *et al.* [47] incorporate a prototype contrastive loss to mitigate the impact of false positive samples, and leverage an imputation strategy based on data specificity to strengthen overall robustness. Yang *et al.* [40] build positive pairs from available data via random cross-view sampling to alleviate false negative effects. Beyond methods centered on isolated sample pairs, Wong *et al.* [35] introduce the concept of neighbor groups to formulate relational contexts. Li *et al.* [14], on the other hand, enhance clustering by directly exploiting existing subgraphs through a three-step process that extracts basic information, separates refined graph structures from biased errors via tensor nuclear norm, and completes them with cross-view learning. In a kernel-based direction, Zhang *et al.* [48] extract low-rank tensor by combining kernel learning and subspace learning, and iteratively impute missing kernels via a self-expressive formulation to improve cluster characterization. He *et al.* [10] bridge inter-view correlations via paired landmarks to overcome information scarcity in incomplete data. Additionally, Xia *et al.* [37] leverage inter-view similarities to exploit spatial structure, and apply connectivity constraints on similarity matrices such that the resulting graph exhibits low-rank structure and accurately reflects relationships among observed data. Despite these advances, the widespread neglect of view-exclusive artifacts during consensus learning could corrupt similarity measures, and inconsistent anchor orders across views disrupt graph construction. Practically, the hyper-parameter reliance also hinders broad deployment.

3. Method

3.1. Preliminaries

In this work, we denote $\{\mathbf{D}_v \in \mathbb{R}^{d_v \times n}\}_{v=1}^{V}$ as an incomplete multi-view dataset consisting of n instances across V views, with index vectors $\{\mathbf{m}_v \in \mathbb{R}^{n_v}\}_{v=1}^{V}$. d_v is the feature dimension and n_v is the number of observed samples on view v. Then, we can build the indicator matrices $\{\mathbf{E}_v \in \mathbb{R}^{n \times n_v}\}_{v=1}^{V}$ where $[\mathbf{E}_v]_{i,j}$ is set to 1 when $[\mathbf{m}_v]_j$ is equal to i otherwise as 0. Accordingly, we can obtain that $\mathbf{D}_v\mathbf{E}_v \in \mathbb{R}^{d_v \times n_v}$ denotes the collection of available samples on view v. On the basis of this, the IMVC learning can be formulated as

$$\begin{aligned}\min_{\mathbf{W}_v,\mathbf{W}} \sum_{v=1}^{V} \|\mathbf{D}_v\mathbf{E}_v - \mathbf{D}_v\mathbf{E}_v\mathbf{W}_v\|_F^2 + \eta\psi\left(\mathbf{W}, \mathbf{E}_v\mathbf{W}_v\mathbf{E}_v^\top\right) \\ \text{s.t. } \mathbf{W}^\top\mathbf{1}_n = \mathbf{1}_n, \mathbf{W} \geq 0, \mathbf{W}_{i,i} = 0,\ i = 1, \cdots, n, \\ \mathbf{W}_v^\top\mathbf{1}_{n_v} = \mathbf{1}_{n_v}, \mathbf{W}_v \geq 0, [\mathbf{W}_v]_{j,j} = 0,\ j = 1, \cdots, n_v,\end{aligned} \tag{1}$$

where $\mathbf{W}_v \in \mathbb{R}^{n_v \times n_v}$ is the self-expression similarity generated on view v, and $\mathbf{W} \in \mathbb{R}^{n \times n}$ is the unified similarity. $\psi(\cdot,\cdot)$ denotes certain fusion schemes, and fuses $\{\mathbf{W}_v\}_{v=1}^{V}$ to generate $\mathbf{W}$. η is the regularization hyper-parameter.

The operation $\mathbf{E}_v\mathbf{W}_v\mathbf{E}_v^\top$ serves to transform the incomplete similarity $\mathbf{W}_v$ with the size of $n_v \times n_v$ into one with the same size as $\mathbf{W}$, thereby enabling the successful aggregation of all view information via fusion. The clustering results are obtained by executing spectral partitioning on $\mathbf{W}$.

3.2. Consensus Learning for IMVC

While the learning paradigm (1) delivers promising results, it generally suffers from at least cubic computational overhead. Moreover, the fusion process may compromise certain similarity components and bring about structural distortions, impairing the clustering performance. In light of these limitations, anchor based consensus learning is developed. It chooses to construct bipartite affinity with small size instead of self-expression matrix with full size, and learns to directly generate a consensus affinity for all views. The framework can be formulated as

$$\begin{aligned}\min_{\mathbf{H}} \sum_{v=1}^{V} \|\mathbf{D}_v\mathbf{E}_v - \mathbf{C}_v\mathbf{H}\mathbf{E}_v\|_F^2 + \eta\phi(\mathbf{H}) \\ \text{s.t. } \mathbf{H}^\top\mathbf{1}_m = \mathbf{1}_n, \mathbf{H} \geq 0,\end{aligned} \tag{2}$$

where $\mathbf{C}_v \in \mathbb{R}^{d_v \times m}$ denotes anchor matrix, constructed by heuristic sampling or random sampling. $\mathbf{H} \in \mathbb{R}^{m \times n}$ denotes consensus affinity. $\phi(\cdot)$ denotes certain regularizers.

By leveraging a small set of anchors to represent the original data, this paradigm constructs a bipartite affinity graph that captures essential similarity patterns. Not only does this strategy decrease computational overhead, but it also mitigates the risk of fusion-induced distortions, leading to more robust and scalable clustering outcomes.

3.3. View-exclusive Artifact Learning

Despite its merits, during the construction of consensus affinity $\mathbf{H}$, (2) does not account for the interference caused by view-exclusive artifacts, which may corrupt the similarity measurement. To this end, we explicitly model the exclusive-to-consensus interplay to decrease such artifact-induced perturbations. Specifically, inspired by (1), we utilize affinity learning conducted on each individual view to characterize view-exclusive artifacts, and then adaptively incorporate it into the consensus learning process. Accordingly, we formulate the framework as

$$\begin{aligned}\min_{\mathbf{H},\mathbf{H}_v,\theta_v} \sum_{v=1}^{V} \|\mathbf{D}_v\mathbf{E}_v - \theta_v\mathbf{C}_v\mathbf{H}\mathbf{E}_v + (\theta_v - 1)\mathbf{C}_v\mathbf{H}_v\mathbf{E}_v\|_F^2 \\ \text{s.t. } 0 \leq \theta_v \leq 1, \mathbf{H}^\top\mathbf{1}_m = \mathbf{1}_n, \mathbf{H} \geq 0, \\ \mathbf{H}_v^\top\mathbf{1}_m = \mathbf{1}_n, \mathbf{H}_v \geq 0,\end{aligned} \tag{3}$$

where $\mathbf{H}_v \in \mathbb{R}^{m\times n}$ denotes the view-exclusive artifact matrix that captures interference unique to view v. θ_v serves as a balancing factor to adaptively regulate $\mathbf{H}$ and $\mathbf{H}_v$. This Janus-faced learning paradigm encourages to extract a more purified and robust consensus affinity.

3.4. Topology Harmonization

In addition to overlooking artifact interference, the anchor order across views in (2) could be inconsistent, which arises from the inherent lack of supervision in clustering setting. Such inconsistencies will cause broken semantic correspondence between views. To remedy this, we introduce a space transformation to harmonize the anchor topology. Specially, building upon (2), we extend the framework as

$$\begin{aligned}\min_{\mathbf{H},\mathbf{Q}_v}\sum_{v=1}^{V}\left\|\mathbf{D}_v\mathbf{E}_v-\mathbf{C}_v\mathbf{Q}_v\mathbf{H}\mathbf{E}_v\right\|_F^2\\ \text{s.t. }\ \mathbf{H}^\top\mathbf{1}_m=\mathbf{1}_n,\mathbf{H}\geq 0,\\ \mathbf{Q}_v^\top\mathbf{1}_m=\mathbf{1}_m,\mathbf{Q}_v\mathbf{1}_m=\mathbf{1}_m,\mathbf{Q}_v\in\{0,1\},\end{aligned} \tag{4}$$

where $\mathbf{Q}_v \in \mathbb{R}^{m\times m}$ with unary encoding constraints serve to synchronize anchor order and meanwhile preserve their original values.

3.5. Framework Overview

Given that different views usually exhibit varying levels of importance or contribution, we associate each view with a weighting coefficient to dynamically calibrate its contribution. In conjunction with (3) and (4), therefore, we design the overall objective function as

$$\begin{aligned}\min_{\mathbf{\Omega}}\sum_{v=1}^{V}\beta_v^2\left\|\mathbf{D}_v\mathbf{E}_v-\theta_v\mathbf{C}_v\mathbf{Q}_v\mathbf{H}\mathbf{E}_v+(\theta_v-1)\mathbf{C}_v\mathbf{H}_v\mathbf{E}_v\right\|_F^2\\ \text{s.t. }\ \boldsymbol{\beta}\geq 0,\boldsymbol{\beta}^\top\mathbf{1}=1,0\leq\theta_v\leq 1,\mathbf{C}_v^\top\mathbf{C}_v=\mathbf{I}_m,\\ \mathbf{H}^\top\mathbf{1}_m=\mathbf{1}_n,\mathbf{H}\geq 0,\mathbf{H}_v^\top\mathbf{1}_m=\mathbf{1}_n,\mathbf{H}_v\geq 0,\\ \mathbf{Q}_v^\top\mathbf{1}_m=\mathbf{1}_m,\mathbf{Q}_v\mathbf{1}_m=\mathbf{1}_m,\mathbf{Q}_v\in\{0,1\},\end{aligned} \tag{5}$$

where $\mathbf{\Omega}=\{\mathbf{C}_v,\mathbf{Q}_v,\mathbf{H},\mathbf{H}_v,\theta_v,\beta_v\}$.

3.6. Optimization Process

We employ an alternating minimization strategy to address the objective (5).

▲ With fixed $\mathbf{Q}_v$, $\mathbf{H}$, $\mathbf{H}_v$, θ_v and β_v, the problem (5) regarding the variable $\mathbf{C}_v$ can be equivalently simplified as

$$\min_{\mathbf{C}_v^\top\mathbf{C}_v=\mathbf{I}_m}\left\|\mathbf{D}_v\mathbf{E}_v-\theta_v\mathbf{C}_v\mathbf{Q}_v\mathbf{H}\mathbf{E}_v+(\theta_v-1)\mathbf{C}_v\mathbf{H}_v\mathbf{E}_v\right\|_F^2 \tag{6}$$

Expanding F-norm through trace operation, we can further transform (6) as

$$\max_{\mathbf{C}_v^\top\mathbf{C}_v=\mathbf{I}_m}\operatorname{Tr}\left(\mathbf{D}_v\mathbf{E}_v\left(\theta_v\mathbf{Q}_v\mathbf{H}\mathbf{E}_v-(\theta_v-1)\mathbf{H}_v\mathbf{E}_v\right)^\top\mathbf{C}_v^\top\right). \tag{7}$$

Denote $\mathbf{U}$ and $\mathbf{V}$ as the singular vector matrices of the item $\mathbf{D}_v\mathbf{E}_v\left(\theta_v\mathbf{Q}_v\mathbf{H}\mathbf{E}_v-(\theta_v-1)\mathbf{H}_v\mathbf{E}_v\right)^\top$, and then we have that the optimal solution of $\mathbf{C}_v$ is $\mathbf{U}\mathbf{V}^\top$.

Remark 1. *Combined with the matrix size, i.e., $\mathbf{D}_v \in \mathbb{R}^{d_v\times n}$, $\mathbf{E}_v \in \mathbb{R}^{n\times n_v}$, $\mathbf{Q}_v \in \mathbb{R}^{m\times m}$, $\mathbf{H} \in \mathbb{R}^{m\times n}$ and $\mathbf{H}_v \in \mathbb{R}^{m\times n}$, we have that constructing this item will require at least $\mathcal{O}(d_vnn_v+m^2n+mnn_v+d_vn_vm)$ computational complexity. Evidently, when the incomplete percentage is low, i.e., n_v is close to n, this complexity will be approach $\mathcal{O}(n^2)$. Such high cost is not beneficial for large-scale deployment. To reduce it to linear, we observe that $\mathbf{D}_v\mathbf{E}_v\left(\theta_v\mathbf{Q}_v\mathbf{H}\mathbf{E}_v-(\theta_v-1)\mathbf{H}_v\mathbf{E}_v\right)^\top$ is equal to $\mathbf{D}_v\mathbf{E}_v\mathbf{E}_v^\top\left(\theta_v\mathbf{Q}_v\mathbf{H}\mathbf{E}_v-(\theta_v-1)\mathbf{H}_v\right)^\top$ and that $\mathbf{E}_v\mathbf{E}_v^\top$ is diagonal. Further, we have $\mathbf{D}_v\mathbf{E}_v\mathbf{E}_v^\top$ can be equivalently expressed as $\mathbf{D}_v\odot\mathbf{B}_v$, where $\mathbf{B}_v$ is $\mathbf{1}_{d_v}\cdot[\sum_{j=1}^{n_v}[\mathbf{E}_v]_{1,j};\cdots;\sum_{j=1}^{n_v}[\mathbf{E}_v]_{n,j}]^\top_n$. Constructing $\mathbf{D}_v\odot\mathbf{B}_v\left(\theta_v\mathbf{Q}_v\mathbf{H}-(\theta_v-1)\mathbf{H}_v\right)^\top$ will take $\mathcal{O}(m^2n+d_vnm)$ cost. Therefore, we can conduct singular value decomposition on $\mathbf{D}_v\odot\mathbf{B}_v\left(\theta_v\mathbf{Q}_v\mathbf{H}-(\theta_v-1)\mathbf{H}_v\right)^\top$ to generate the singular vector matrices $\mathbf{U}$ and $\mathbf{V}$.*

On the basis of **Remark** 1, we have that the total computing cost about optimizing $\mathbf{C}_v$ is $\mathcal{O}(m^2n+d_vnm+d_vm^2)$, which is linear to n since d_v is a constant and m is generally largely smaller than n.

▲ With fixed $\mathbf{C}_v$, $\mathbf{H}$, $\mathbf{H}_v$, θ_v and β_v, the problem (5) regarding the variable $\mathbf{Q}_v$ can be transformed as

$$\begin{aligned}\max_{\mathbf{Q}_v}\operatorname{Tr}\left(\mathbf{H}\mathbf{E}_v\left(\mathbf{D}_v\mathbf{E}_v+(\theta_v-1)\mathbf{C}_v\mathbf{H}_v\mathbf{E}_v\right)^\top\mathbf{C}_v\mathbf{Q}_v\right)\\ \text{s.t. }\ \mathbf{Q}_v^\top\mathbf{1}_m=\mathbf{1}_m,\mathbf{Q}_v\mathbf{1}_m=\mathbf{1}_m,\mathbf{Q}_v\in\{0,1\}.\end{aligned} \tag{8}$$

By introducing the matrix-vectorization operation Vec, we can equivalently transform the objective of (8) as

$$\max_{\mathbf{Q}_v}\left(\operatorname{Vec}\left(\left(\mathbf{H}\mathbf{E}_v\left(\widehat{\mathbf{G}}_v+\widehat{\mathbf{A}}_v\right)^\top\mathbf{C}_v\right)^\top\right)\right)^\top\operatorname{Vec}\left(\mathbf{Q}_v\right), \tag{9}$$

where $\widehat{\mathbf{G}}_v=\mathbf{D}_v\mathbf{E}_v$ and $\widehat{\mathbf{A}}_v=(\theta_v-1)\mathbf{C}_v\mathbf{H}_v\mathbf{E}_v$.

Remark 2. *In conjunction with the feasible domain, after matrix-vectorization transformation, we have that this problem is expressed as a binary programming, and accordingly can be solved efficiently by existing software packages.*

During optimizing (9), it needs to build the item $\mathbf{H}\mathbf{E}_v\left(\widehat{\mathbf{G}}_v+\widehat{\mathbf{A}}_v\right)^\top\mathbf{C}_v$, which requires $\mathcal{O}(d_vnn_v+d_vmn+mnn_v+mn_vd_v+m^2d_v)$ computing cost. This is close to $\mathcal{O}(n^2)$. To decrease it, we transform (9) as

$$\max_{\mathbf{Q}_v}\left(\operatorname{Vec}\left(\left(\mathbf{C}_v^\top\mathbf{D}_v\mathbf{E}_v\mathbf{E}_v^\top+\widehat{\mathbf{B}}_v\mathbf{E}_v^\top\right)\mathbf{H}^\top\right)\right)^\top\operatorname{Vec}\left(\mathbf{Q}_v\right), \tag{10}$$

where $\widehat{\mathbf{B}}_v=(\theta_v-1)\mathbf{H}_v\mathbf{E}_v$.

Remark 3. *Following the skill handling the item* $\mathbf{D}_v\mathbf{E}_v\mathbf{E}_v^\top$ *in optimizing* $\mathbf{C}_v$*, we can express* $\mathbf{H}_v\mathbf{E}_v\mathbf{E}_v^\top$ *as* $\mathbf{H}_v \odot \mathbf{G}_v$*.* $\mathbf{G}_v$ *is equal to* $\mathbf{1}_m \cdot [\sum_{j=1}^{n_v} [\mathbf{E}_v]_{1,j}; \cdots; \sum_{j=1}^{n_v} [\mathbf{E}_v]_{n,j}]_n^\top$*.*

Therefore, we can further transform (10) as

$$\max_{\mathbf{Q}_v} \left(\mathrm{Vec}\left(\left(\mathbf{C}_v^\top (\mathbf{D}_v \odot \mathbf{B}_v) + \widehat{\mathbf{C}}_v \right) \mathbf{H}^\top \right) \right)^\top \mathrm{Vec}\left(\mathbf{Q}_v\right). \tag{11}$$

where $\widehat{\mathbf{C}}_v = (\theta_v - 1)(\mathbf{H}_v \odot \mathbf{G}_v)$.

Building $\left(\mathbf{C}_v^\top (\mathbf{D}_v \odot \mathbf{B}_v) + (\theta_v - 1)(\mathbf{H}_v \odot \mathbf{G}_v)\right)\mathbf{H}^\top$ only needs $\mathcal{O}(md_vn + m^2n)$ computing overhead. Therefore, the computational complexity of updating $\mathbf{Q}_v$ is $\mathcal{O}(n)$.

▲ With fixed $\mathbf{C}_v$, $\mathbf{Q}_v$, $\mathbf{H}_v$, θ_v and β_v, the problem (5) regarding the variable $\mathbf{H}$ can be expressed as

$$\begin{aligned} \min_{\mathbf{H}} \sum_{v=1}^{V} \beta_v^2 \,\mathrm{Tr}\left(\mathbf{H}^\top \theta_v^2 \mathbf{H}\mathbf{E}_v\mathbf{E}_v^\top - 2\theta_v \mathbf{Q}_v^\top \left(\widehat{\mathbf{J}}_v + \widehat{\mathbf{C}}_v \right) \mathbf{H}^\top \right) \\ \text{s.t. } \mathbf{H}^\top \mathbf{1}_m = \mathbf{1}_n, \mathbf{H} \geq 0, \end{aligned} \tag{12}$$

where $\widehat{\mathbf{J}}_v = \mathbf{C}_v^\top (\mathbf{D}_v \odot \mathbf{B}_v)$.

Leveraging the characteristics inherent to feasible domain, we can optimize $\mathbf{H}$ by column. Thus, we can transform (12) as

$$\begin{aligned} \min_{\mathbf{H}_{:,j}} \mathbf{H}_{:,j}^\top \sum_{v=1}^{V} \beta_v^2 \theta_v^2 \sum_{i=1}^{n_v} [\mathbf{E}_v]_{j,i} \mathbf{H}_{:,j} - 2\mathbf{H}_{:,j}^\top \left[\sum_{v=1}^{V} \beta_v^2 \theta_v \widehat{\mathbf{M}}_v \right]_{:,j} \\ \text{s.t. } \mathbf{H}_{:,j}^\top \mathbf{1}_m = 1, \mathbf{H}_{:,j} \geq 0, \end{aligned} \tag{13}$$

where $\widehat{\mathbf{M}}_v = \mathbf{Q}_v^\top \left(\widehat{\mathbf{J}}_v + \widehat{\mathbf{C}}_v \right)$.

Remark 4. *Due to the incompleteness, the coefficient* $\sum_{v=1}^{V} \beta_v^2 \theta_v^2 \sum_{i=1}^{n_v} [\mathbf{E}_v]_{j,i}$ *may be zero in some scenarios. Accordingly, the objective in (13) becomes linear if it is zero, and quadratic otherwise.*

When $\sum_{v=1}^{V} \beta_v^2 \theta_v^2 \sum_{i=1}^{n_v} [\mathbf{E}_v]_{j,i} \neq 0$, we can equivalently transform (13) as

$$\begin{aligned} \min_{\mathbf{H}_{:,j}} \left\| \mathbf{H}_{:,j} - \frac{\left[\sum_{v=1}^{V} \beta_v^2 \theta_v \widehat{\mathbf{M}}_v \right]_{:,j}}{\sum_{v=1}^{V} \beta_v^2 \theta_v^2 \sum_{i=1}^{n_v} [\mathbf{E}_v]_{j,i}} \right\|_F^2 \\ \text{s.t. } \mathbf{H}_{:,j}^\top \mathbf{1}_m = 1, \mathbf{H}_{:,j} \geq 0. \end{aligned} \tag{14}$$

To solve it, we write its Lagrange function as

$$\mathcal{L} = \frac{1}{2} \left\| \mathbf{H}_{:,j} - \mathbf{s}_j \right\|_F^2 - \mathbf{H}_{:,j}^\top \boldsymbol{\phi} - \sigma \left(\mathbf{H}_{:,j}^\top \mathbf{1}_m - 1 \right), \tag{15}$$

where $\mathbf{s}_j = \frac{\left[\sum_{v=1}^{V} \beta_v^2 \theta_v \widehat{\mathbf{M}}_v \right]_{:,j}}{\sum_{v=1}^{V} \beta_v^2 \theta_v^2 \sum_{i=1}^{n_v} [\mathbf{E}_v]_{j,i}}$. Then, we can get

$$\mathbf{H}_{:,j} = \mathbf{s}_j + \boldsymbol{\phi} + \sigma \mathbf{1}_m, \quad \mathbf{H}_{:,j} \odot \boldsymbol{\phi} = \mathbf{0}. \tag{16}$$

Combined with the column sum constraint, we can further get

$$\mathbf{s}_j^\top \mathbf{1} + m\sigma = 1. \tag{17}$$

Thus, we have

$$\mathbf{H}_{:,j} = \max\left(\frac{1 - \mathbf{1}^\top \mathbf{s}_j}{m} \mathbf{1} + \mathbf{s}_j, 0 \right). \tag{18}$$

When $\sum_{v=1}^{V} \beta_v^2 \theta_v^2 \sum_{i=1}^{n_v} [\mathbf{E}_v]_{j,i} = 0$, (13) equivalently becomes

$$\min_{\mathbf{H}_{:,j}^\top \mathbf{1}_m = 1, \mathbf{H}_{:,j} \geq 0} -\mathbf{H}_{:,j}^\top \left[\sum_{v=1}^{V} \beta_v^2 \theta_v \mathbf{Q}_v^\top \widehat{\mathbf{M}}_v \right]_{:,j}, \tag{19}$$

which is a linear programming problem, and can be easily addressed by existing software.

When the coefficient $\sum_{v=1}^{V} \beta_v^2 \theta_v^2 \sum_{i=1}^{n_v} [\mathbf{E}_v]_{j,i} \neq 0$, the problem (13) has a closed-form solution. It only needs to construct the vector $\mathbf{s}_j$, which takes $\mathcal{O}(md_vn + m^2n)$ computing cost. When $\sum_{v=1}^{V} \beta_v^2 \theta_v^2 \sum_{i=1}^{n_v} [\mathbf{E}_v]_{j,i} = 0$, it needs $\mathcal{O}(m^3n)$ cost via linear programming to update $\mathbf{H}$. Therefore, the computational complexity of updating $\mathbf{H}$ is $\mathcal{O}(n)$.

▲ With fixed $\mathbf{C}_v$, $\mathbf{Q}_v$, $\mathbf{H}$, θ_v and β_v, the problem (5) regarding the variable $\mathbf{H}_v$ can equivalently become

$$\begin{aligned} \min_{\mathbf{H}_v} \mathrm{Tr}\Big((\theta_v - 1)^2 \mathbf{H}_v^\top \mathbf{H}_v \mathbf{E}_v \mathbf{E}_v^\top + 2(\theta_v - 1) \cdot \\ \mathbf{H}_v^\top \left(\mathbf{C}_v^\top \mathbf{D}_v \mathbf{E}_v \mathbf{E}_v^\top - \theta_v \mathbf{Q}_v \mathbf{H}\mathbf{E}_v \mathbf{E}_v^\top \right) \Big) \\ \text{s.t. } \mathbf{H}_v^\top \mathbf{1}_m = \mathbf{1}_n, \mathbf{H}_v \geq 0. \end{aligned} \tag{20}$$

Remark 5. *Constructing the item* $\mathbf{Q}_v\mathbf{H}\mathbf{E}_v\mathbf{E}_v^\top$ *takes* $\mathcal{O}(m^2n + mn^2)$ *computing overhead. To reduce it, with the strategy tackling the matrix product* $\mathbf{D}_v\mathbf{E}_v\mathbf{E}_v^\top$ *in optimizing* $\mathbf{C}_v$*, we can equivalently express* $\mathbf{H}\mathbf{E}_v\mathbf{E}_v^\top$ *as* $\mathbf{H} \odot \mathbf{G}_v$*, where* $\mathbf{G}_v$ *has been constructed in optimizing* $\mathbf{Q}_v$*.*

Together with the column sum constraints of $\mathbf{H}_v$ and the unary encoding characteristics of $\mathbf{E}_v$, we can further transform (20) as

$$\begin{aligned} \min_{[\mathbf{H}_v]_{:,j}} [\mathbf{H}_v]_{:,j}^\top (\theta_v - 1)^2 \sum_{i=1}^{n_v} [\mathbf{E}_v]_{j,i} [\mathbf{H}_v]_{:,j} + 2(\theta_v - 1) \cdot \\ [\mathbf{H}_v]_{:,j}^\top \left[\mathbf{C}_v^\top (\mathbf{D}_v \odot \mathbf{B}_v) - \theta_v \mathbf{Q}_v (\mathbf{H} \odot \mathbf{G}_v) \right]_{:,j} \\ \text{s.t. } [\mathbf{H}_v]_{:,j}^\top \mathbf{1}_m = 1, [\mathbf{H}_v]_{:,j} \geq 0. \end{aligned} \tag{21}$$

When the coefficient $(\theta_v - 1)^2 \sum_{i=1}^{n_v} [\mathbf{E}_v]_{j,i}$ is non-zero, in conjunction with the skill tackling (13), we can derive

$$[\mathbf{H}_v]_{:,j} = \max\left(\frac{1 + \mathbf{1}^\top \mathbf{t}_j^v}{m} \mathbf{1} - \mathbf{t}_j^v, 0 \right), \tag{22}$$

where $\mathbf{t}_j^v = \frac{\left[\mathbf{C}_v^\top(\mathbf{D}_v \odot \mathbf{B}_v) - \theta_v \mathbf{Q}_v(\mathbf{H} \odot \mathbf{G}_v)\right]_{:,j}}{(\theta_v - 1)\sum_{i=1}^{n_v}[\mathbf{E}_v]_{j,i}}$.

When $(\theta_v - 1)^2 \sum_{i=1}^{n_v}[\mathbf{E}_v]_{j,i}$ is zero, (21) becomes a linear programming, and can be solved via existing software.

The construction of $\mathbf{C}_v^\top(\mathbf{D}_v \odot \mathbf{B}_v)$ and $\mathbf{Q}_v(\mathbf{H} \odot \mathbf{G}_v)$ takes $\mathcal{O}(md_vn)$ and $\mathcal{O}(m^2n)$ overhead, respectively. The linear programming takes $\mathcal{O}(m^3n)$ overhead. Therefore, updating $\mathbf{H}_v$ requires $\mathcal{O}(n)$ computational complexity.

▲ With fixed $\mathbf{C}_v$, $\mathbf{Q}_v$, $\mathbf{H}$, $\mathbf{H}_v$ and β_v, the problem (5) regarding the variable θ_v equivalently reduces to

$$\min_{0\le\theta_v\le 1} \operatorname{Tr}\left(\theta_v^2\left(\widehat{\mathbf{L}}_v - \widehat{\mathbf{P}}_v\right)\cdot\left(\widehat{\mathbf{L}}_v - \widehat{\mathbf{P}}_v\right)^\top + 2\theta_v\left(\widehat{\mathbf{L}}_v - \widehat{\mathbf{P}}_v\right)\left(\mathbf{D}_v\mathbf{E}_v - \widehat{\mathbf{L}}_v\right)^\top\right), \tag{23}$$

where $\widehat{\mathbf{L}}_v = \mathbf{C}_v\mathbf{H}_v\mathbf{E}_v$, $\widehat{\mathbf{P}}_v = \mathbf{C}_v\mathbf{Q}_v\mathbf{H}\mathbf{E}_v$.

Remark 6. *The objective in (23) is a quadratic function about θ_v, and consequently, without considering constraints, the optimal solution is*

$$\frac{\operatorname{Tr}\left(\left(\widehat{\mathbf{L}}_v - \mathbf{C}_v\mathbf{Q}_v\mathbf{H}\mathbf{E}_v\right)\left(\widehat{\mathbf{L}}_v - \mathbf{D}_v\mathbf{E}_v\right)^\top\right)}{\operatorname{Tr}\left(\left(\widehat{\mathbf{L}}_v - \mathbf{C}_v\mathbf{Q}_v\mathbf{H}\mathbf{E}_v\right)\left(\widehat{\mathbf{L}}_v - \mathbf{C}_v\mathbf{Q}_v\mathbf{H}\mathbf{E}_v\right)^\top\right)}. \tag{24}$$

With **Remark** 6, building $\mathbf{C}_v\mathbf{H}_v\mathbf{E}_v - \mathbf{C}_v\mathbf{Q}_v\mathbf{H}\mathbf{E}_v$ and $\mathbf{C}_v\mathbf{H}_v\mathbf{E}_v - \mathbf{D}_v\mathbf{E}_v$ takes $\mathcal{O}(d_vm^2 + d_vmn + d_vnn_v)$ and $\mathcal{O}(d_vmn + d_vnn_v)$ computing overhead, respectively. Evidently, it is close to a quadratic cost with respect to the sample size n.

To decrease it, we can equivalently convert the numerator and denominator of (24), respectively, to

$$\operatorname{Tr}\left(\left(\mathbf{H}_v \odot \mathbf{G}_v - \mathbf{C}_v^\top(\mathbf{D}_v \odot \mathbf{B}_v)\right)(\mathbf{H}_v - \mathbf{Q}_v\mathbf{H})^\top\right)$$

and

$$\operatorname{Tr}\left(\left(\mathbf{H}_v \odot \mathbf{G}_v - \mathbf{Q}_v(\mathbf{H} \odot \mathbf{G}_v)\right)(\mathbf{H}_v - \mathbf{Q}_v\mathbf{H})^\top\right). \tag{25}$$

Building $\left(\mathbf{H}_v \odot \mathbf{G}_v - \mathbf{C}_v^\top(\mathbf{D}_v \odot \mathbf{B}_v)\right)(\mathbf{H}_v - \mathbf{Q}_v\mathbf{H})^\top$ and $\left(\mathbf{H}_v \odot \mathbf{G}_v - \mathbf{Q}_v(\mathbf{H} \odot \mathbf{G}_v)\right)(\mathbf{H}_v - \mathbf{Q}_v\mathbf{H})^\top$ takes $\mathcal{O}(md_vn + m^2n)$ and $\mathcal{O}(m^2n)$ computing overhead respectively, both of which are linear to the sample size. Denote $\widehat{\mathbf{K}}_v = \mathbf{H}_v \odot \mathbf{G}_v$ and $\widehat{\mathbf{E}}_v = \mathbf{H}_v - \mathbf{Q}_v\mathbf{H}$. Combined with the constraint conditions on θ_v, thus, we have

$$\theta_v = \operatorname{clip}\left(\frac{\operatorname{Tr}\left(\left(\widehat{\mathbf{K}}_v - \mathbf{C}_v^\top(\mathbf{D}_v \odot \mathbf{B}_v)\right)\widehat{\mathbf{E}}_v^\top\right)}{\operatorname{Tr}\left(\left(\widehat{\mathbf{K}}_v - \mathbf{Q}_v(\mathbf{H} \odot \mathbf{G}_v)\right)\widehat{\mathbf{E}}_v^\top\right)}, 0, 1\right). \tag{26}$$

▲ With fixed $\mathbf{C}_v$, $\mathbf{Q}_v$, $\mathbf{H}$, $\mathbf{H}_v$ and θ_v, the problem (5) regarding the variable β_v becomes

$$\min_{\boldsymbol{\beta}\ge 0, \boldsymbol{\beta}^\top\mathbf{1}=1} \sum_{v=1}^{V} \beta_v^2 q_v. \tag{27}$$

Algorithm 1 The Proposed PJFTH Algorithm

Input: Dataset $\{\mathbf{D}_v\}_{v=1}^V$, index vectors $\{\mathbf{m}_v\}_{v=1}^V$,
Initialize: $\mathbf{C}_v$, $\mathbf{Q}_v$, $\mathbf{H}$, $\mathbf{H}_v$, θ_v, β_v;

repeat
  Update $\mathbf{C}_v$ via (7);
  Update $\mathbf{Q}_v$ via (11);
  Update $\mathbf{H}$ via (18) or (19);
  Update $\mathbf{H}_v$ via (21);
  Update θ_v via (26);
  Update β_v via (28);
until convergent

Output: Spectral clustering on $\mathbf{H}$;

q_v is $\|\mathbf{D}_v\mathbf{E}_v - \theta_v\mathbf{C}_v\mathbf{Q}_v\mathbf{H}\mathbf{E}_v + (\theta_v - 1)\mathbf{C}_v\mathbf{H}_v\mathbf{E}_v\|_F^2$, a constant about β_v. With Cauchy inequality, we can obtain the optimal solution of β_v is $\frac{1}{q_v}/\sum_{v=1}^V \frac{1}{q_v}$.

Remark 7. *Constructing q_v needs $\mathcal{O}(d_vnn_v + d_vm^2 + d_vmn)$ computing overhead, which is close to $\mathcal{O}(n^2)$. Noticing that each column of $\mathbf{E}_v$ consists of a single 1 and all other entries are 0, we can derive that $\|\mathbf{D}_v\mathbf{E}_v\|_F^2$ is equivalent to $\left\|\mathbf{D}_v\mathbf{E}_v\mathbf{E}_v^\top\right\|_F^2$. Further, we have that q_v is equivalent to $\left\|\mathbf{D}_v\mathbf{E}_v\mathbf{E}_v^\top - \theta_v\mathbf{C}_v\mathbf{Q}_v\mathbf{H}\mathbf{E}_v\mathbf{E}_v^\top + (\theta_v - 1)\mathbf{C}_v\mathbf{H}_v\mathbf{E}_v\mathbf{E}_v^\top\right\|_F^2$*

On the basis of **Remark** 7, in conjunction with the strategies handling $\mathbf{D}_v$, $\mathbf{H}$, $\mathbf{H}_v$ with $\mathbf{E}_v\mathbf{E}_v^\top$, we can derive that q_v can be equivalently expressed as $\left\|\mathbf{D}_v \odot \mathbf{B}_v - \theta_v\mathbf{C}_v\mathbf{Q}_v(\mathbf{H} \odot \mathbf{G}_v) + \widehat{\mathbf{F}}_v\right\|_F^2$, where $\widehat{\mathbf{F}}_v = (\theta_v - 1)\mathbf{C}_v(\mathbf{H}_v \odot \mathbf{G}_v)$. Therefore, we have

$$\beta_v = \frac{\frac{1}{\left\|\mathbf{D}_v \odot \mathbf{B}_v - \theta_v\mathbf{C}_v\mathbf{Q}_v(\mathbf{H} \odot \mathbf{G}_v) + \widehat{\mathbf{F}}_v\right\|_F^2}}{\sum_{v=1}^V \frac{1}{\left\|\mathbf{D}_v \odot \mathbf{B}_v - \theta_v\mathbf{C}_v\mathbf{Q}_v(\mathbf{H} \odot \mathbf{G}_v) + \widehat{\mathbf{F}}_v\right\|_F^2}}. \tag{28}$$

Constructing $\left\|\mathbf{D}_v \odot \mathbf{B}_v - \theta_v\mathbf{C}_v\mathbf{Q}_v(\mathbf{H} \odot \mathbf{G}_v) + \widehat{\mathbf{F}}_v\right\|_F^2$ requires $\mathcal{O}(d_vm^2 + d_vmn)$ computing overhead. Besides, the data dimension d_v is irrelevant to n, and m is largely smaller than n. As a result, updating β_v takes $\mathcal{O}(dmn)$ overhead totally where d is equal to $\sum_{v=1}^V d_v$. That is, the computational complexity of optimizing β_v is $\mathcal{O}(n)$.

Algorithm 1 outlines the complete workflow of PJFTH.

Remark 8. *The overall computational complexity of our PJFTH is dominated by the updates of $\mathbf{C}_v$, $\mathbf{Q}_v$, $\mathbf{H}$, $\mathbf{H}_v$, θ_v and β_v, and accordingly is linear about the sample size n.*

Remark 9. *The overall space complexity of PJFTH is also linear since during updating each variable, it only involves linear storage overhead.*

Table 1. Clustering Results Comparison. The best results are highlighted in black.

Dataset		WASHING									ORLR								
Missing Ratio		30%			60%			80%			30%			60%			80%		
Method	NH	ACC	PUR	FSC	ACC	PUR	FSC	ACC	PUR	FSC	ACC	PUR	FSC	ACC	PUR	FSC	ACC	PUR	FSC
HCMSC	3	40.87	50.83	39.24	30.43	49.13	29.25	30.00	46.96	28.54	30.39	33.25	10.13	24.55	26.33	7.64	20.21	21.45	5.09
LSIVC	2	47.39	48.26	48.47	45.82	48.01	48.53	45.27	47.89	47.83	13.52	12.25	14.85	13.05	12.13	14.17	13.11	12.02	13.93
LRIVC	2	36.87	50.43	34.77	26.78	47.72	26.36	29.28	48.57	27.40	39.80	42.51	24.44	35.83	38.31	18.98	35.18	37.60	18.58
AGIMC	2	44.35	47.39	46.67	42.61	46.52	45.39	41.30	46.52	44.18	42.25	44.50	10.91	27.25	30.00	5.66	30.50	33.75	6.58
IVCBG	1	49.45	43.11	32.10	45.64	40.01	24.08	43.66	38.36	22.66	51.79	55.03	**43.32**	43.70	46.70	34.67	32.56	34.65	28.21
PIMVC	2	42.11	46.52	44.56	43.04	48.22	44.87	43.89	47.72	45.70	36.66	40.44	18.82	30.58	33.83	13.91	29.51	32.46	13.36
GRIMC	3	27.24	46.65	25.55	28.93	47.07	26.36	25.37	46.52	24.84	16.43	17.31	2.36	29.63	31.65	12.75	22.21	23.58	6.37
UIMC	0	27.39	46.52	24.45	28.26	46.01	25.16	25.65	46.25	24.64	18.50	19.25	3.28	18.00	19.00	3.30	16.00	16.75	2.22
OSIMC	1	53.48	**61.61**	44.83	48.26	57.30	40.09	48.70	**55.43**	40.16	50.75	55.25	34.95	40.50	43.75	23.11	32.25	35.00	15.68
USETL	2	43.24	47.33	44.54	45.59	47.96	46.82	46.24	48.41	47.03	31.83	37.84	7.98	31.74	36.29	6.59	28.49	32.49	5.63
LBIVC	1	32.17	46.52	29.95	31.30	47.39	29.41	41.74	48.70	36.30	33.50	36.25	16.81	31.00	32.75	13.85	35.00	36.50	17.34
HCLGL	1	47.83	51.30	47.20	47.14	51.47	47.11	46.92	50.67	46.53	42.54	46.66	10.64	30.53	35.54	7.16	29.91	34.18	6.62
Ours	0	**53.91**	59.52	**49.81**	**52.91**	**57.93**	**49.31**	**50.13**	54.13	**48.44**	**57.13**	**60.31**	42.47	**52.31**	**55.36**	**35.84**	**44.51**	**47.69**	**28.33**
		NUSCENE									NUJECTEN								
HCMSC	3	9.36	31.38	6.42	9.48	31.74	6.53	9.21	31.20	6.36	21.56	29.31	16.06	22.40	29.70	15.70	20.79	28.82	15.30
LSIVC	2	11.28	33.90	7.30	11.43	**33.16**	7.21	10.33	32.61	6.81	25.08	31.23	19.48	22.40	30.21	19.32	22.37	30.14	18.40
LRIVC	2	11.20	**34.16**	6.90	9.80	32.68	6.50	9.44	32.36	6.40	25.03	32.17	19.26	23.83	**31.03**	19.63	22.22	30.37	18.64
AGIMC	2	12.24	31.43	9.28	12.55	31.53	10.12	12.72	31.74	10.09	22.13	30.02	15.57	20.98	28.32	14.42	19.42	29.42	15.43
IVCBG	1	6.53	3.98	9.77	6.40	3.91	9.50	6.35	3.87	9.13	22.59	31.96	16.36	22.05	30.00	15.79	21.04	30.53	15.63
PIMVC	2	10.71	34.02	6.88	10.65	33.02	6.80	10.24	**33.41**	6.72	24.29	31.07	18.13	23.98	30.92	18.20	23.48	**30.88**	17.75
GRIMC	3	11.24	33.32	7.13	11.55	32.87	7.47	10.61	32.62	7.19	25.03	**32.23**	18.51	22.47	30.87	17.28	22.66	30.03	17.39
UIMC	0	7.13	30.13	5.89	6.92	31.47	5.73	7.02	30.05	5.93	18.39	24.13	18.01	17.87	24.35	17.43	18.24	23.81	17.12
OSIMC	1	10.82	32.77	7.64	9.16	31.09	6.41	8.35	30.28	6.09	21.88	31.34	16.83	20.27	30.14	15.18	18.91	30.00	15.00
USETL	2	11.57	29.48	11.59	12.49	29.50	11.02	10.58	29.78	10.61	18.12	25.56	16.30	18.93	25.37	18.76	20.09	25.46	20.25
LBIVC	1	9.74	30.55	6.79	9.26	30.84	6.78	9.30	30.89	6.54	18.59	29.00	15.10	18.11	29.15	15.22	18.56	29.10	14.60
HCLGL	1	12.02	33.58	7.53	11.29	32.94	7.47	11.59	32.20	7.32	21.93	30.68	17.14	20.42	30.19	16.66	19.74	29.57	16.98
Ours	0	**14.02**	30.12	**11.66**	**13.87**	30.00	**11.55**	**13.73**	29.82	**11.63**	**25.16**	27.82	**23.71**	**24.33**	26.23	**23.66**	**23.74**	26.52	**22.86**
		NUSWDBJ									YOUBESEL								
HCMSC	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
LSIVC	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
LRIVC	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
AGIMC	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
IVCBG	1	12.69	18.90	7.78	11.73	18.14	7.30	11.78	21.10	7.23	16.89	**26.85**	8.23	14.90	25.39	8.13	13.74	25.79	8.20
PIMVC	2	-	-	-	**13.80**	**23.11**	9.30	12.79	**22.94**	8.73	-	-	-	-	-	-	-	-	-
GRIMC	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
UIMC	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
OSIMC	1	8.20	14.97	5.83	8.36	15.36	5.48	7.90	15.72	5.20	10.38	26.62	7.35	13.34	26.43	10.59	15.24	26.12	12.37
USETL	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
LBIVC	1	8.91	18.53	6.63	8.59	18.17	6.38	8.76	17.97	6.07	-	-	-	-	-	-	-	-	-
HCLGL	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ours	0	**14.36**	**19.36**	**10.28**	12.53	16.84	**9.85**	**13.04**	17.79	**9.88**	**18.16**	26.83	**13.78**	**18.67**	**26.68**	**14.53**	**17.38**	**26.63**	**13.48**

4. Experiments and Analysis

4.1. Comparison Methods and Benchmark Datasets

To evaluate the clustering performance, six datasets are used in experiments. Their details are listed in Table 2. The following twelve algorithms serve as comparison algorithms: HCMSC [16], LSIVC [22], LRIVC [3], AGIMC [33], IVCBG [30], PIMVC [4], GRIMC [14], UIMC [20], OSIMC [51], USETL [1], LBIVC [32], and HCLGL [31]. We organize experiments under multiple missing ratios, including 30%, 60%, 80%. The clustering results are presented in Table 1 where NH means the number of hyper-parameters.

Table 2. Details of Datasets

Dataset	Samples	Views	Dimensions	Clusters
WASHING	230	4	230, 230, 1703, 230	5
ORLR	400	2	288, 1024	40
NUSCENE	4095	5	128, 73, 225, 64, 144	33
NUJECTEN	6251	5	129, 145, 74, 65, 226	10
NUSWDBJ	30000	5	65, 145, 226, 129, 74	31
YOUBESEL	101499	5	64, 647, 838, 64,512	31

4.2. Observations

Several observations can be drawn from Table 1:
1. Our method produces superior results in most cases, for example on ORLR and YOUBESEL, which suggests that the proposed PJFTH is effective in addressing IMVC tasks.
2. Against LSIVC, AGIMC, GRIMC that solve IMVC tasks by subspace or kernel, our anchor-based scheme makes better results, illustrating that our anchor strategy is working.
3. For suboptimal results, a potential reason is that we first formulate the spectral embedding and then perform partitioning rather than directly deriving discrete cluster labels.
More observations are in supplementary material, Section 5.

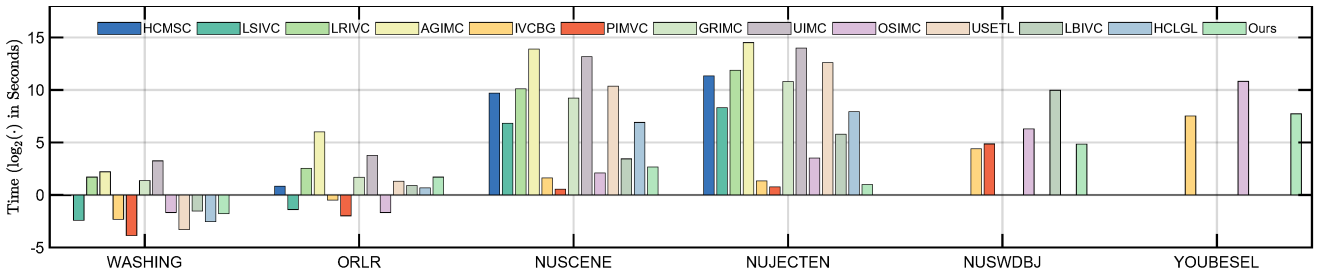
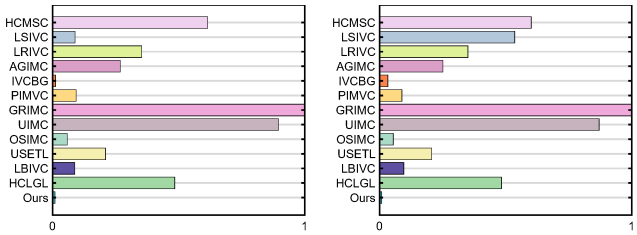
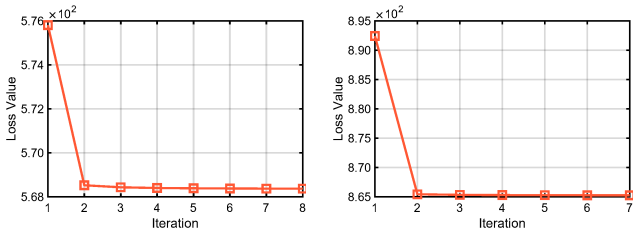


Figure 2. Running Time



(a) NUSCENE (b) NUJECTEN

Figure 3. Memory Overhead Comparison



(a) NUSCENE (b) NUJECTEN

Figure 4. Loss Value Evolution

Table 3. Ablation Study

Ablation	WASHING			ORLR			NUSCENE		
	ACC	PUR	FSC	ACC	PUR	FSC	ACC	PUR	FSC
OVAL	50.35	54.80	48.56	51.45	54.55	35.02	13.51	29.41	10.99
OTH	50.43	54.59	48.54	51.59	53.94	35.27	13.12	29.23	11.01
Ours	**52.91**	**57.93**	**49.31**	**52.31**	**55.36**	**35.84**	**13.87**	**30.00**	**11.55**
	NUJECTEN			NUSWDBJ			YOUBESEL		
OVAL	23.22	25.32	22.43	11.60	16.28	9.15	17.96	26.12	14.37
OTH	23.68	24.34	23.29	11.15	16.04	9.24	18.21	25.93	14.29
Ours	**24.33**	**26.23**	**23.66**	**12.53**	**16.84**	**9.85**	**18.67**	**26.68**	**14.53**

4.3. Resource Overhead Comparison

Time Overhead. Fig. 2 illustrates that in most cases, we require less runtime, revealing the efficiency of our method. Besides, we can execute on all these datasets, while methods like HCMSC, LSIVC, LRIVC, etc, do not have this ability. Accordingly, our method has a wider applicability.

Memory Overhead. Fig. 3 shows that we achieve a notable reduction in space storage against several competitors, which mainly benefits from the used bipartite graph scheme.

4.4. Ablation Study

The ablation results are reported in Table 3 where OVAL and OTH denote not involving view-exclusive artifact learning (VAL) and topology harmonization (TH), respectively. As seen, after VAL and TH, the performance is improved.

4.5. Convergence

Fig. 4 displays that with increasing iterations, the objective value gradually declines and eventually approaches a stable state, which illustrates the convergence property of PJFTH.

5. Limitations

The orthogonal constraint helps improve the discriminability of the learned anchors, while it may cause anchors to deviate from the space characteristics of the original data. Constructing the anchors that have similar distributions to original data would further enhance clustering performance. Besides, the two-stage pipeline that first builds affinity matrix and then conducts spectral partitioning could exacerbate clustering variance. Directly formulating cluster indicators from original data is a promising way to boost the stability.

6. Conclusion

In the work, an IMVC method named PJFTH is specifically designed to alleviate key challenges including view-exclusive artifact interference, anchor order inconsistency, regularization hyper-parameter reliance. It introduces a Janus-faced affinity learning mechanism that explicitly separates and suppresses the artifact interference, and employs a permutation transformation with unary encoding constraints to harmonize anchor topology and meanwhile maintain the originally learned anchor values. The entire pipeline operates without any hyper-parameters, offering a practical plug-and-play functionality. Moreover, all components are jointly optimized under a unified objective function, and the linear complexity ensures high scalability and usability. Experimental evaluations from multiple perspectives show the effectiveness and competitiveness of PJFTH.

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